

计算机网络

实验三 数据包抓取与分析

计算机与软件学院
深圳大学



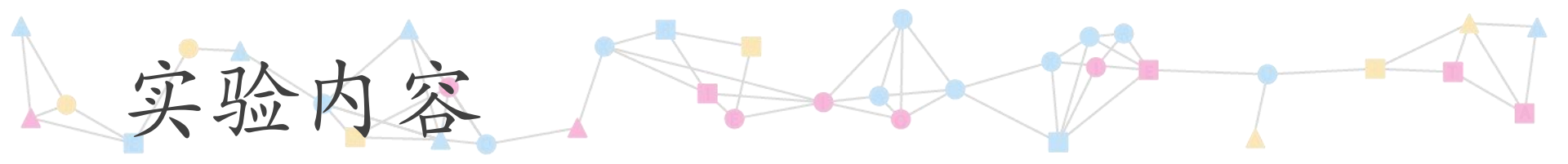
实验三 数据包抓取与分析

■ 实验目的

- 学习安装、使用协议分析软件，掌握基本的数据报抓取、过滤和分析方法，能分析HTTP、TCP、ICMP等协议。

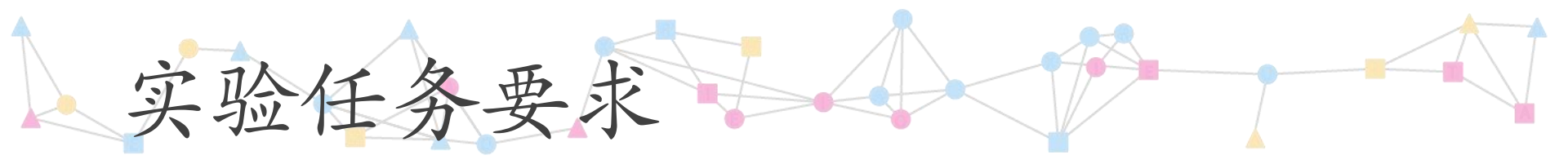
■ 实验环境

- 使用具有Internet连接的Windows操作系统；
- 抓包软件Wireshark。



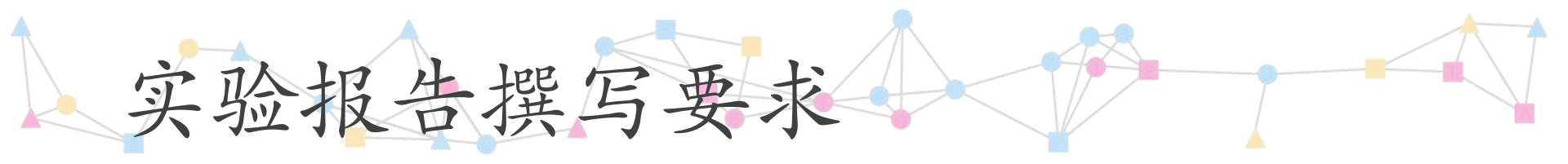
实验内容

1. 安装学习Wireshark软件
2. 抓包与分析HTTP协议
3. 分析TCP协议
4. 分析TCP三次握手



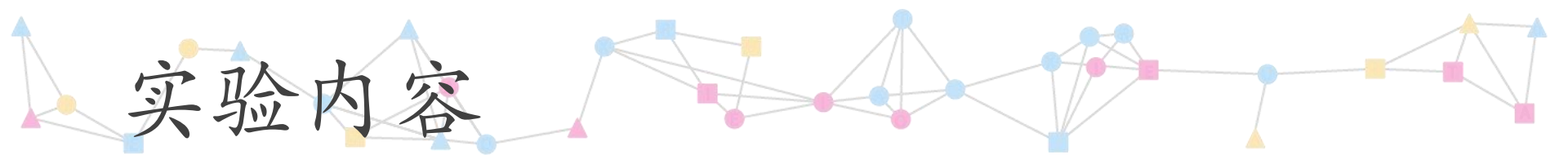
实验任务要求

- 请参考本讲义学习Wireshark软件的使用方法
- 安装Wireshark软件
- 理解TCP/IP协议分层模型
- 了解TCP报文格式
- 理解TCP三次握手
- 依照步骤完成实验内容1—5
- 对实验结果截图
- 撰写实验报告



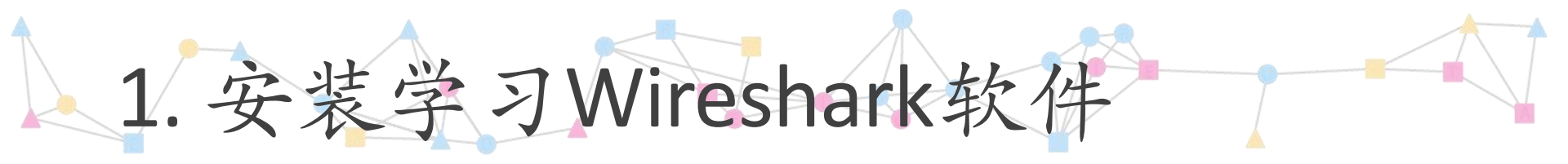
实验报告撰写要求

- 使用教务处制作的实验报告模板
- 注意按进度填写实验时间和实验报告提交时间
- 填写模板中的每一部分
- 填写实验步骤时，做到条理清晰
- 注意截图清晰、美观
- 要求在演示操作步骤的截图上加标注，指出操作步骤和操作结果，没有会被扣分
- 实验报告只有截图，没有文字说明讲解会扣分
- 实验结果要有原理分析，否则会被扣分
- 出现一模一样的实验报告，均得零分



实验内容

1. 安装学习Wireshark软件
2. 抓包与分析HTTP协议
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1. 安装学习Wireshark软件

- Wireshark是世界上最广泛使用的网络协议分析器。
- 官网地址：

<https://www.wireshark.org/>

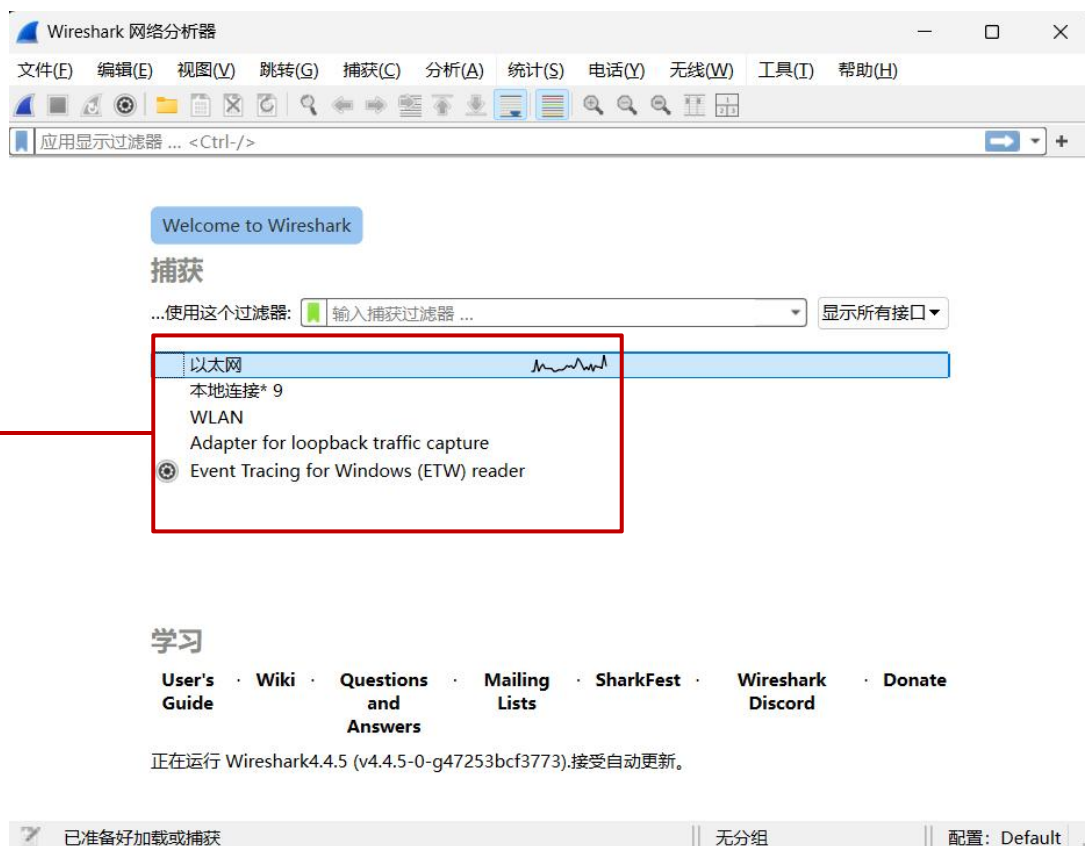
- 软件使用手册：

https://www.wireshark.org/docs/wsug_html_chunked

1) 请下载并安装该软件。(本讲义Wireshark版本号为4.4.5)

1. 安装学习Wireshark软件

- 2) 运行Wireshark，初始界面如下图。
- 3) 从接口列表中选择要捕获的接口，双击即可开始捕获。

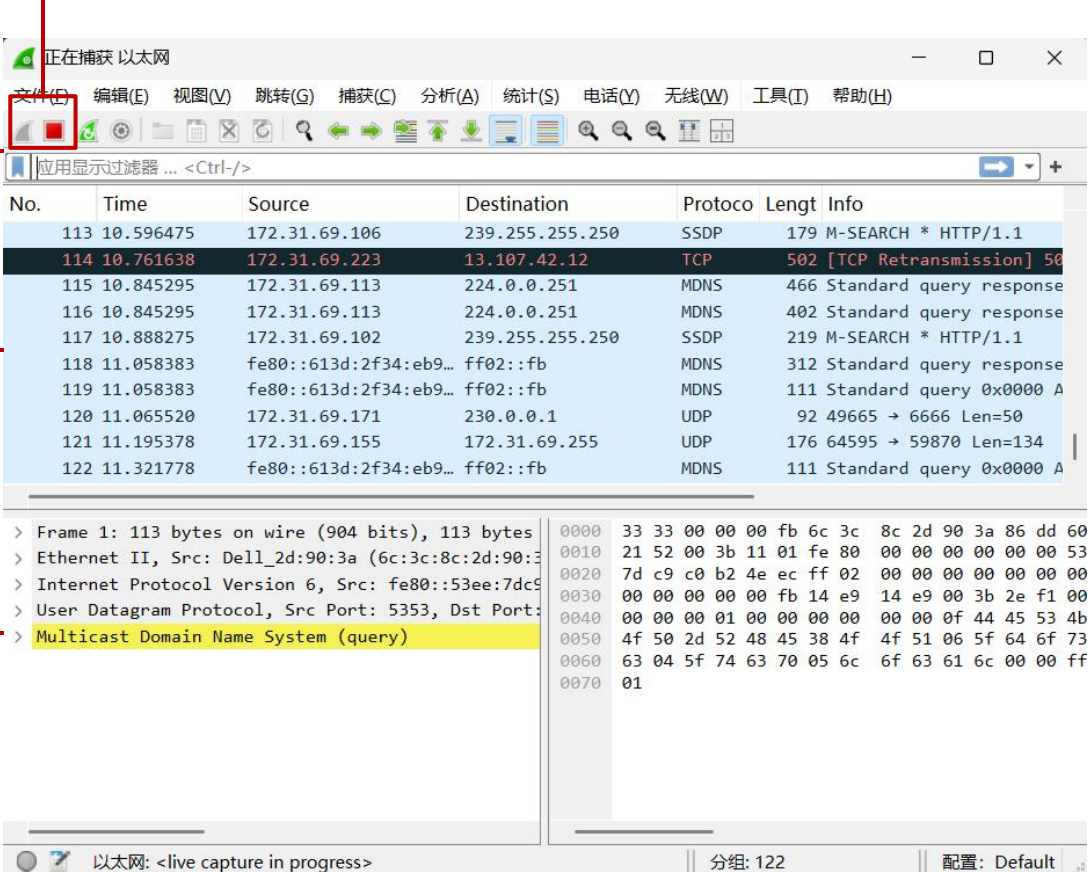


接口列表

1. 安装学习Wireshark软件

4) Wireshark进入主界面，并开始捕获分组。

开始/停止捕获分组



The screenshot shows the Wireshark network protocol analyzer interface. The title bar reads '正在捕获 以太网'. The menu bar includes '文件(F)', '编辑(E)', '视图(V)', '跳转(G)', '捕获(C)', '分析(A)', '统计(S)', '电话(Y)', '无线(W)', '工具(I)', and '帮助(H)'. The toolbar contains various icons for file operations, capture control, and analysis. A red box highlights the '开始/停止捕获' (Start/Stop Capture) icon. Below the toolbar is the '过滤器' (Filter) field with the text '应用显示过滤器 ... <Ctrl-/>'. The main pane is divided into two sections: the '分组列表栏' (Packet List Pane) on the left and the '分组字节栏' (Packet Bytes Pane) on the right. The packet list shows a table of captured packets with columns for 'No.', 'Time', 'Source', 'Destination', 'Protocol', 'Length', and 'Info'. Packet 114 is highlighted in red. The packet bytes pane shows the raw data of the selected packet in hexadecimal and ASCII. The status bar at the bottom indicates '以太网: <live capture in progress>', '分组: 122', and '配置: Default'.

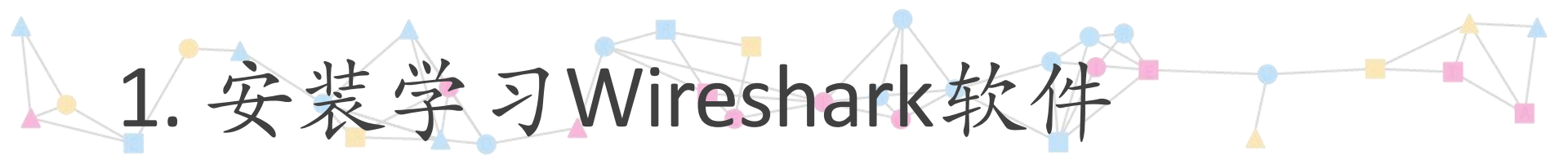
过滤器

分组列表栏

分组详情栏

状态栏

分组字节栏



1. 安装学习Wireshark软件

5) 学习使用过滤器?

- 协议过滤

- 举例: http

- IP地址过滤

- 举例: `ip.src == 192.168.2.178 and ip.dst == 184.86.198.104`
 - 含义: 过滤源地址是192.168.2.178并且目的地址是184.86.198.104的分组

- 模式过滤

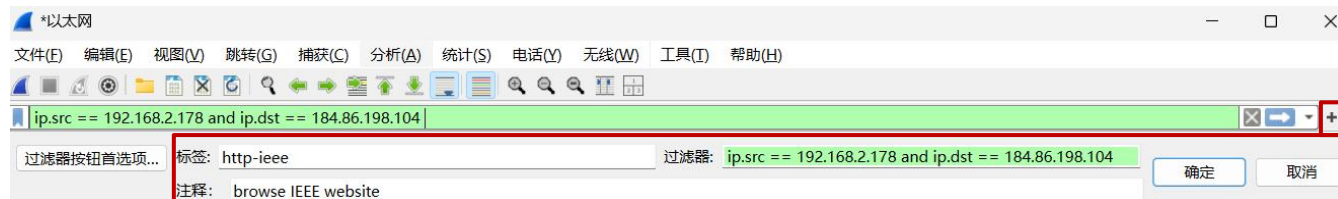
- 举例: `http.request.method=="GET"`
 - 含义: 过滤http请求方法是GET的分组

- 端口过滤

- 举例: `tcp.port == 80`
 - 含义: 过滤tcp端口号是80的分组

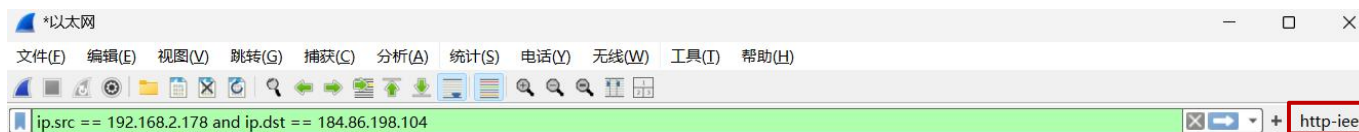
1. 安装学习Wireshark软件

6) 添加一个显示过滤器按钮



1. 添加一个显示过滤器按钮

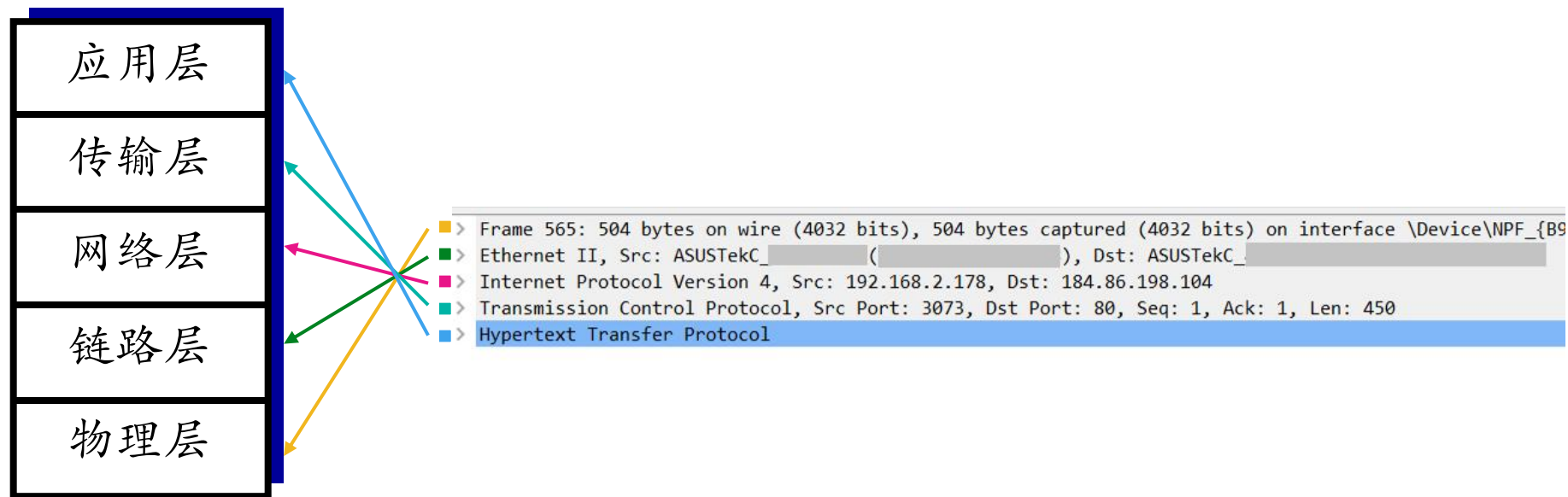
2. 填写信息

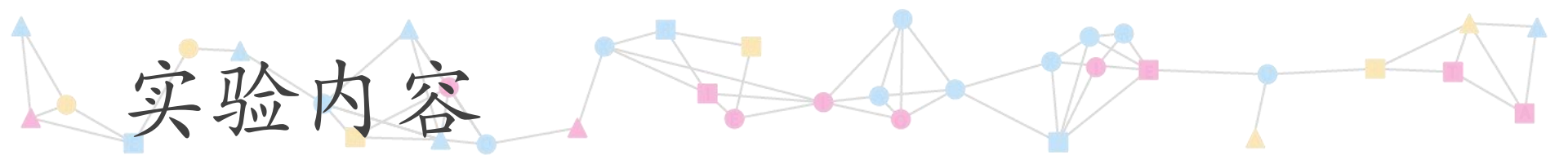


3. 按钮添加成功

1. 安装学习Wireshark软件

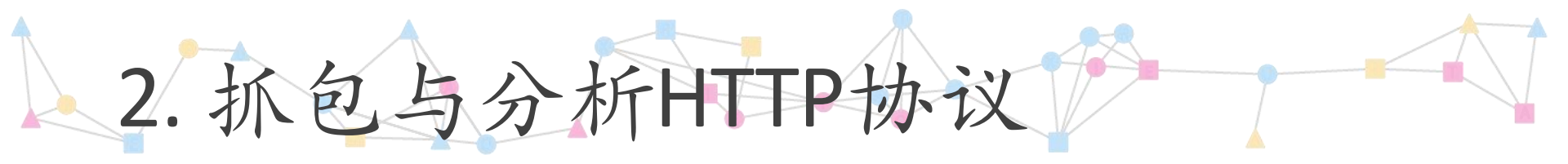
6) 分组详情栏





实验内容

1. 安装学习Wireshark软件
2. 抓包与分析HTTP协议
3. 分析TCP协议
4. 分析TCP三次握手



2. 抓包与分析HTTP协议

- 1) 开启Wireshark抓包，在过滤器中输入http，即过滤http协议的分组。
- 2) 打开浏览器，输入一个网址（例如 ieeexplore.ieee.org）。注意：为了避免浏览器缓存起作用，最好使用chrome浏览器的incognito隐身模式。

2. 抓包与分析HTTP协议

3) 观察到Wireshark分组列表栏中出现了HTTP协议分组。

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文件(E) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(I) 帮助(H)

http www.szu.edu.cn

No.	Time	Source	Destination	Protocol	Length	Info
23368	183.354574	20.219.237.180	172.31.69.223	HTTP	523	HTTP/1.1 302 Found
23388	183.767661	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
23396	183.786398	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
24049	183.895241	172.31.247.35	172.31.69.223	HTTP	1468	HTTP/1.1 206 Partial Content
24057	183.898562	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
24960	184.012703	172.31.247.35	172.31.69.223	HTTP	290	HTTP/1.1 206 Partial Content
24995	184.015916	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
26025	184.101796	172.31.247.35	172.31.69.223	HTTP	290	HTTP/1.1 206 Partial Content
26041	184.105294	172.31.69.223	172.31.247.35	HTTP	512	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
27163	184.209962	172.31.247.35	172.31.69.223	HTTP	290	HTTP/1.1 206 Partial Content
28184	184.300364	172.31.247.35	172.31.69.223	HTTP	346	HTTP/1.1 206 Partial Content

> Frame 23368: 523 bytes on wire (4184 bits), 523 bytes captured (4184 bits) on interface \Device\NPF_{E9395...}

> Ethernet II, Src: HuaweiTechno_9c:71:27 (1c:e6:39:9c:71:27), Dst: HP_cf:c6:15 (7c:57:58:cf:c6:15)

> Internet Protocol Version 4, Src: 20.219.237.180, Dst: 172.31.69.223

> Transmission Control Protocol, Src Port: 80, Dst Port: 52370, Seq: 1, Ack: 434, Len: 469

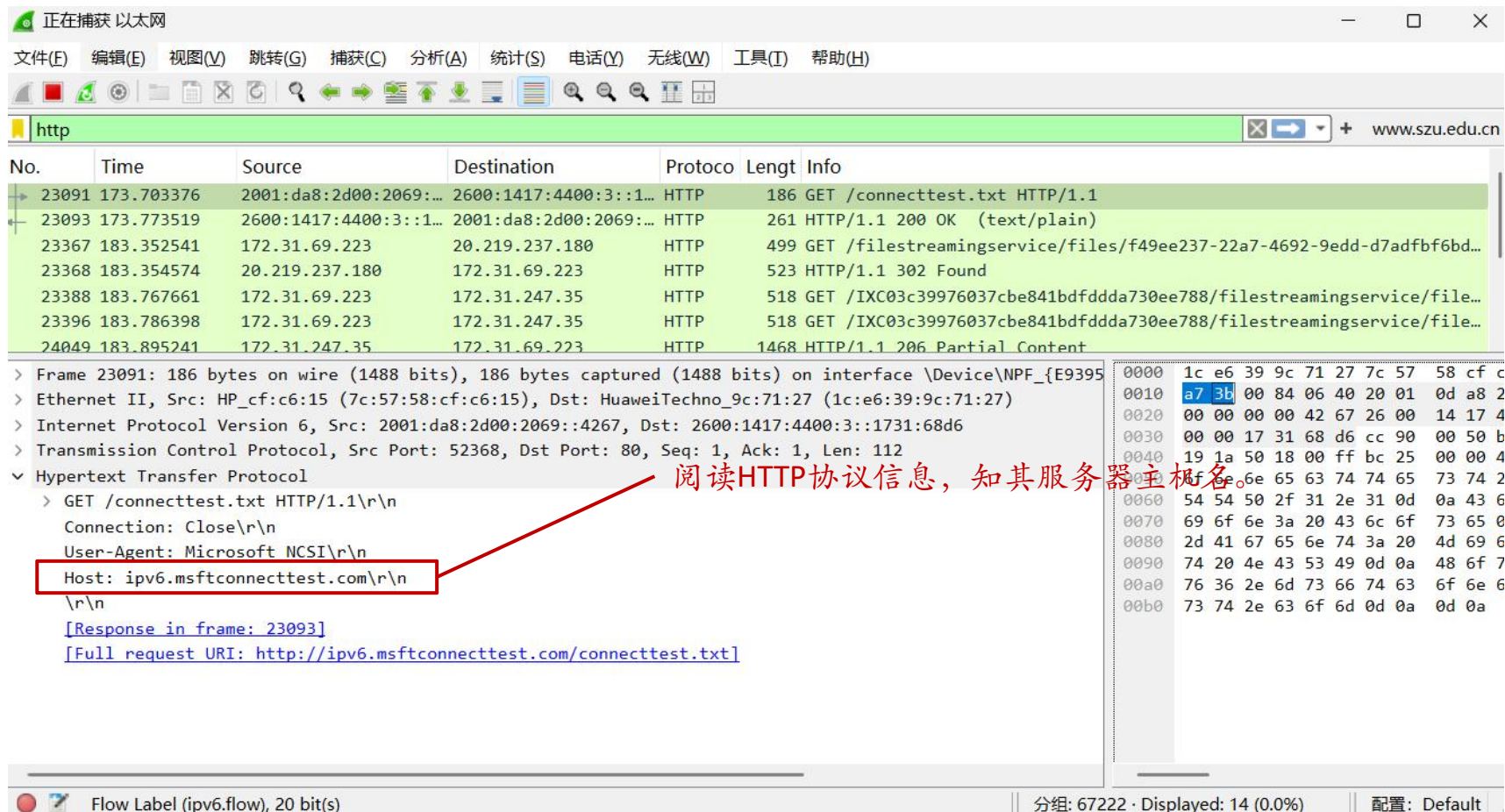
> Hypertext Transfer Protocol

0000	7c 57 58 cf c6 15 1c e6 39 9c
0010	01 fd a8 0e 40 00 7a 06 62 5e
0020	45 df 00 50 cc 92 ed 97 75 39
0030	00 0a 7f 9e 00 00 48 54 54 50
0040	30 32 20 46 6f 75 6e 64 0d 0a
0050	3a 20 6e 67 69 6e 78 0d 0a 41
0060	43 6f 6e 74 72 6f 6c 2d 41 6c
0070	65 64 65 6e 74 69 61 6c 73 3a
0080	0a 41 63 63 65 73 73 2d 43 6f
0090	41 6c 6c 6f 77 2d 4f 72 69 67
00a0	0a 41 63 63 65 73 73 2d 43 6f
00b0	45 78 70 6f 73 65 2d 48 65 61
00c0	4c 6f 63 61 74 69 6f 6e 2c 20

Hypertext Transfer Protocol: Protocol 分组: 30511 · Displayed: 14 (0.0%) 配置: Default

2. 抓包与分析HTTP协议

- 4) 分析哪些分组是前一步浏览网页发生的。单击任意分组，分组详情栏显示其具体信息。



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文件(E) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(I) 帮助(H)

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No.	Time	Source	Destination	Protocol	Length	Info
23091	173.703376	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	HTTP	186	GET /connecttest.txt HTTP/1.1
23093	173.773519	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	HTTP	261	HTTP/1.1 200 OK (text/plain)
23367	183.352541	172.31.69.223	20.219.237.180	HTTP	499	GET /filestreamingservice/files/f49ee237-22a7-4692-9edd-d7adfbf6bd...
23368	183.354574	20.219.237.180	172.31.69.223	HTTP	523	HTTP/1.1 302 Found
23388	183.767661	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
23396	183.786398	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
24049	183.895241	172.31.247.35	172.31.69.223	HTTP	1468	HTTP/1.1 206 Partial Content

> Frame 23091: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface \Device\NPF_{E9395...}

> Ethernet II, Src: HP_cf:c6:15 (7c:57:58:cf:c6:15), Dst: HuaweiTechno_9c:71:27 (1c:e6:39:9c:71:27)

> Internet Protocol Version 6, Src: 2001:da8:2d00:2069::4267, Dst: 2600:1417:4400:3::1731:68d6

> Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 112

✓ Hypertext Transfer Protocol

> GET /connecttest.txt HTTP/1.1\r\n

Connection: Close\r\n

User-Agent: Microsoft NCSE\r\n

Host: ipv6.msftconnecttest.com\r\n

\r\n

[Response in frame: 23093]

[Full request URI: http://ipv6.msftconnecttest.com/connecttest.txt]

Flow Label (ipv6.flow), 20 bit(s)

分组: 67222 · Displayed: 14 (0.0%)

配置: Default

2. 抓包与分析HTTP协议

- 5) 从步骤四所得的分组，获知此次通信的源IP地址和目的IP地址。(这里，2001:da8:2d00:2069::4267是私有IP地址，所以是用户的主机。)

正在捕获 以太网

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(I) 帮助(H)

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No.	Time	Source	Destination	Protocol	Length	Info
23091	173.703376	2001:da8:2d00:2069::4267	2600:1417:4400:3::1	HTTP	186	GET /connecttest.txt HTTP/1.1
23093	173.773519	2600:1417:4400:3::1	2001:da8:2d00:2069::4267	HTTP	261	HTTP/1.1 200 OK (text/plain)
23367	183.352541	172.31.69.223	20.219.237.180	HTTP	499	GET /filestreamingservice/files/f49ee237-22a7-4692-9edd-d7adfbf6bd...
23368	183.354574	20.219.237.180	172.31.69.223	HTTP	523	HTTP/1.1 302 Found
23388	183.767661	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
23396	183.786398	172.31.69.223	172.31.247.35	HTTP	518	GET /IXC03c39976037cbe841bdfddda730ee788/filestreamingservice/file...
24049	183.895241	172.31.247.35	172.31.69.223	HTTP	1468	HTTP/1.1 206 Partial Content

> Frame 23091: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface \Device\NPF_{E9395...}

> Ethernet II, Src: HP_cf:c6:15 (7c:57:58:cf:c6:15), Dst: HuaweiTechno_9c:71:27 (1c:e6:39:9c:71:27)

> Internet Protocol Version 6, Src: 2001:da8:2d00:2069::4267, Dst: 2600:1417:4400:3::1731:68d6

> Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 112

✓ Hypertext Transfer Protocol

> GET /connecttest.txt HTTP/1.1\r\n

Connection: Close\r\n

User-Agent: Microsoft NCSI\r\n

Host: ipv6.msftconnecttest.com\r\n\r\n

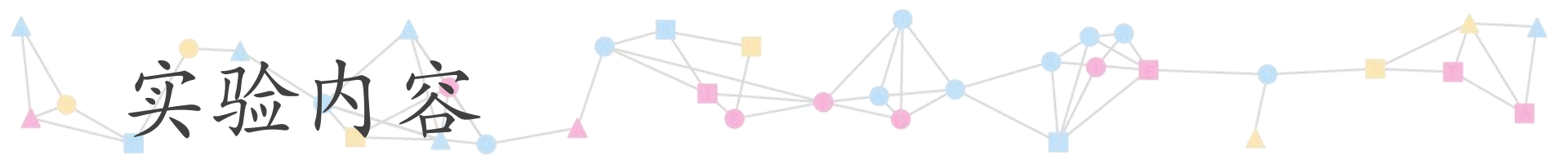
[Response in frame: 23093]

[Full request URI: http://ipv6.msftconnecttest.com/connecttest.txt]

0000 1c e6 39 9c 71 27 7c 57 58 cf c
0010 a7 3b 00 84 06 40 20 01 0d a8 2
0020 00 00 00 00 42 67 26 00 14 17 4
0030 00 00 17 31 68 d6 cc 90 00 50 b
0040 19 1a 50 18 00 ff bc 25 00 00 4
0050 6f 6e 6e 65 63 74 74 65 73 74 2
0060 54 54 50 2f 31 2e 31 0d 0a 43 6
0070 69 6f 6e 3a 20 43 6c 6f 73 65 0
0080 2d 41 67 65 6e 74 3a 20 4d 69 6
0090 74 20 4e 43 53 49 0d 0a 48 6f 7
00a0 76 36 2e 6d 73 66 74 63 6f 6e 6
00b0 73 74 2e 63 6f 6d 0d 0a 0d 0a

Flow Label (ipv6.flow), 20 bit(s) 分组: 67222 · Displayed: 14 (0.0%) 配置: Default

阅读HTTP协议信息，知其服务器主机名。



实验内容

1. 安装学习Wireshark软件
2. 抓包与分析HTTP协议
3. 分析TCP协议
4. 分析TCP三次握手
5. 分析ICMP协议

3.分析 TCP协议

1) 对2-4的分组，分析TCP协议信息。

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文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

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No.	Time	Source	Destination	Protocol	Length	Info
23091	173.703376	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	HTTP	186	GET /connecttest.txt HTTP/1.1

Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 112

Source Port: 52368

Destination Port: 80 源端口号和目的端口号

[Stream index: 223]

[Stream Packet Number: 4]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 112]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3158848141 序列号

[Next Sequence Number: 113 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 2302613786 确认号

0101 = Header Length: 20 bytes (5) 报头长度

Flags: 0x018 (PSH, ACK) 标志位

Window: 255 窗口大小

[Calculated window size: 65280]

[Window size scaling factor: 256]

Checksum: 0xbc25 [unverified] 校验和

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

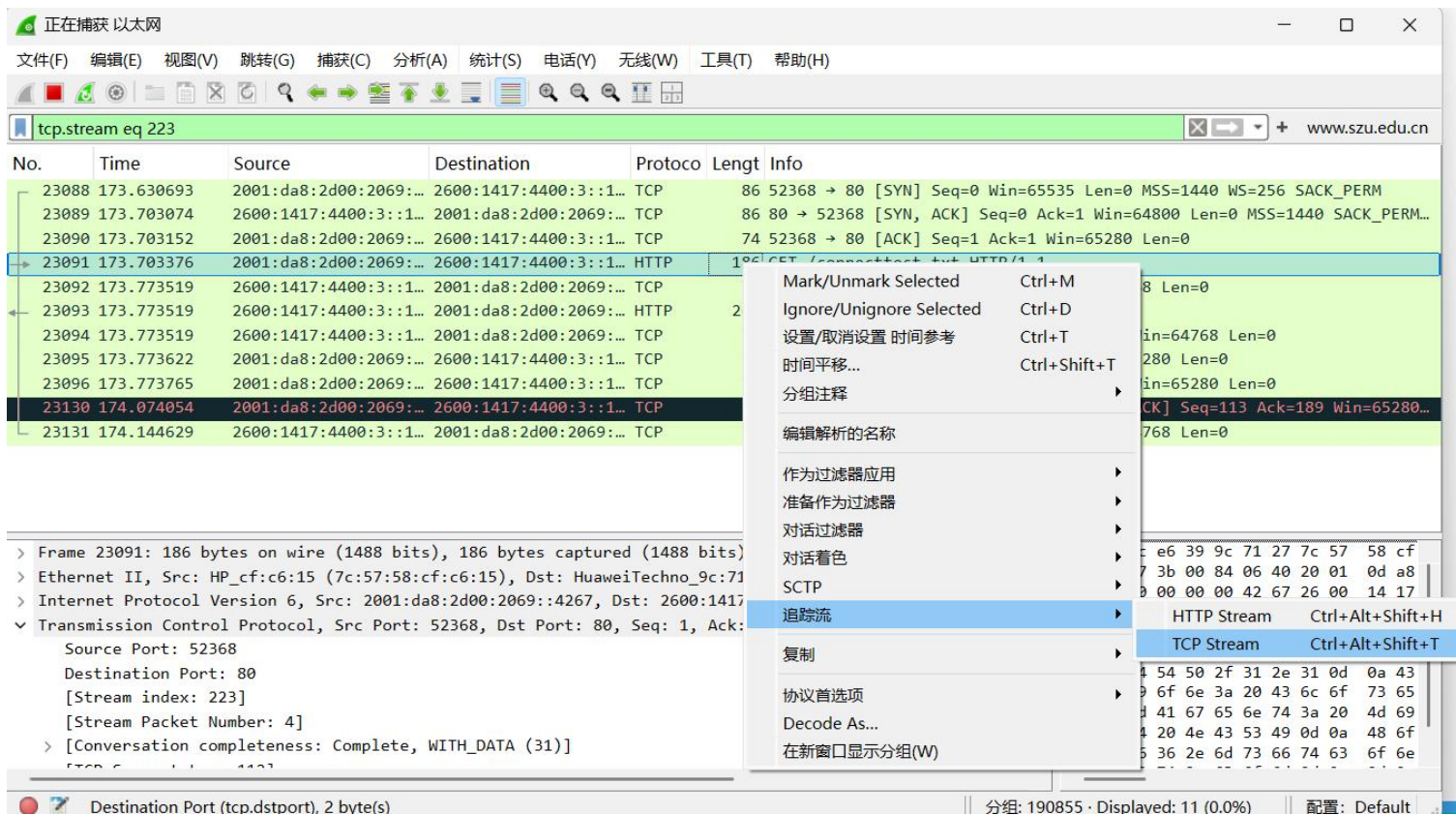
TCP payload (112 bytes) 数据

0000 1c e6 39 9c 71 27 7c 57 58 cf c6
0010 a7 3b 00 84 06 40 20 01 0d a8 2c
0020 00 00 00 00 42 67 26 00 14 17 44
0030 00 00 17 31 68 d6 cc 90 00 50 bc
0040 19 1a 50 18 00 ff bc 25 00 00 47
0050 6f 6e 6e 65 63 74 74 65 73 74 2e
0060 54 54 50 2f 31 2e 31 0d 0a 43 6f
0070 69 6f 6e 3a 20 43 6c 6f 73 65 0c
0080 2d 41 67 65 6e 74 3a 20 4d 69 63
0090 74 20 4e 43 53 49 0d 0a 48 6f 73
00a0 76 36 2e 6d 73 66 74 63 6f 6e 6e
00b0 73 74 2e 63 6f 6d 0d 0a 0d 0a

Destination Port (tcp.dstport), 2 byte(s) 分组: 187093 · Displayed: 24 (0.0%) 配置: Default

3.分析 TCP协议

2) 对2-4的分组，追踪其TCP流。点击右键，从下拉菜单中选择TCP流。



3.分析 TCP协议

3) 对2-4的分组，追踪其TCP流。点击右键，从下拉菜单中选择TCP流。

The image shows the Wireshark network traffic analysis interface. The top bar indicates "正在捕获 以太网" (Capturing on Ethernet). The menu bar includes File (F), Edit (E), View (V), Go (G), Capture (C), Analyze (A), Statistics (S), Tools (T), and Help (H). The toolbar contains various icons for file operations, capture control, and analysis. The packet list pane shows a list of captured packets. Packet 23091 is selected, and its details are expanded in the packet details pane. The packet details pane shows the following information:

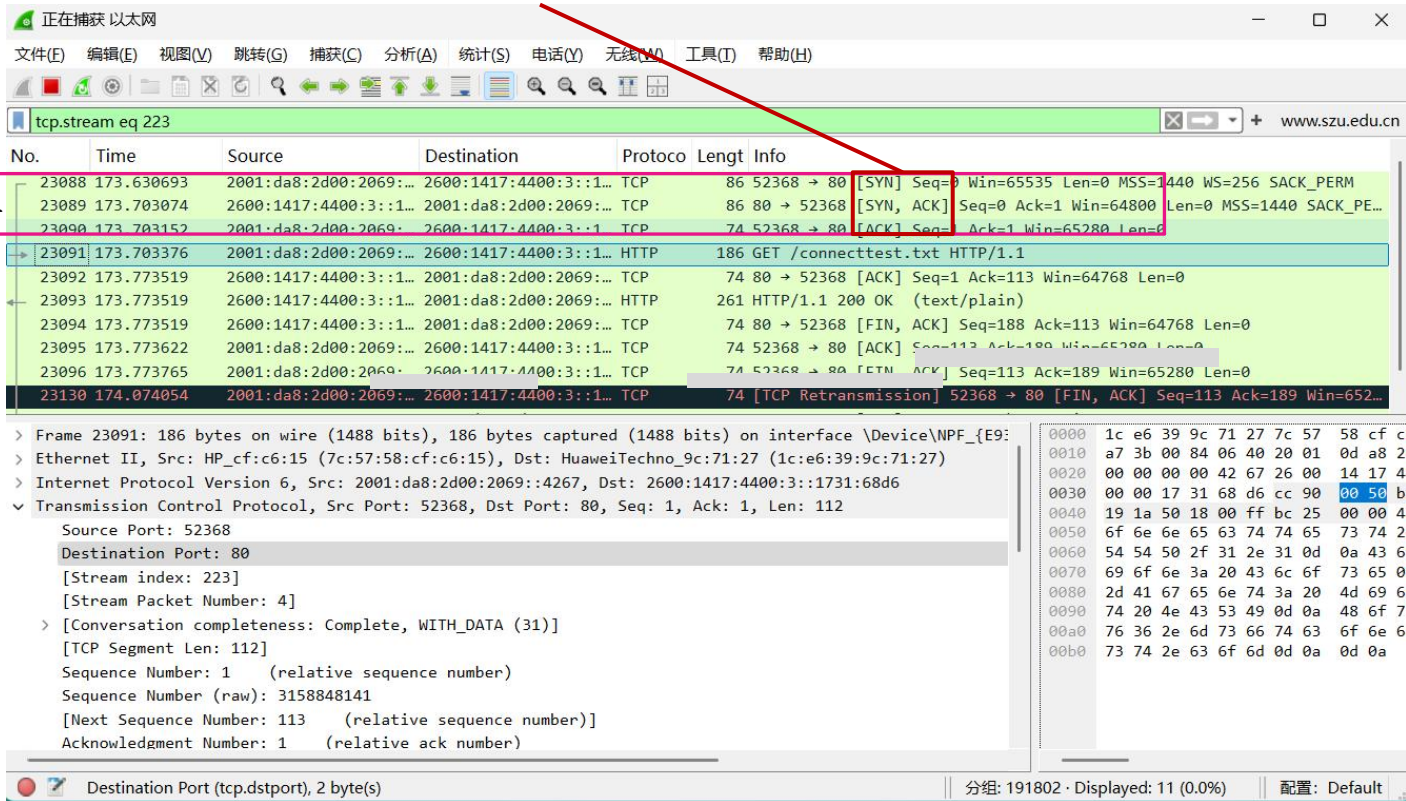
- Frame 23091: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface \Device\NPF_{E9:71:27:1C:E6:39:9C:71:27}
- Ethernet II, Src: HP_cf:c6:15 (7c:57:58:cf:c6:15), Dst: HuaweiTechno_9c:71:27 (1c:e6:39:9c:71:27)
- Internet Protocol Version 6, Src: 2001:da8:2d00:2069::4267, Dst: 2600:1417:4400:3::1731:68d6
- Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 112
- Source Port: 52368
- Destination Port: 80
- [Stream index: 223]
- [Stream Packet Number: 4]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 112]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 3158848141
- [Next Sequence Number: 113 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)

The packet bytes pane shows the raw data of the selected packet, displayed in hexadecimal and ASCII. The status bar at the bottom indicates "Destination Port (tcp.dstport), 2 byte(s)" and "分组: 191802 · Displayed: 11 (0.0%)".

3.分析 TCP协议

- 4) 找到TCP建立连接的分组。原理：1) TCP连接建立应该在HTTP GET请求之前完成; 2) TCP 建立连接时会设置标志位SYN。

TCP 连接建立的分组



正在捕获 以太网

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(I) 帮助(H)

tcp.stream eq 223

No.	Time	Source	Destination	Protocol	Length	Info
23088	173.630693	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	86	52368 → 80 [SYN, Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM...
23089	173.703074	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	TCP	86	80 → 52368 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PE...
23090	173.703152	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
23091	173.703376	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	HTTP	186	GET /connecttest.txt HTTP/1.1
23092	173.773519	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	TCP	74	80 → 52368 [ACK] Seq=1 Ack=113 Win=64768 Len=0
23093	173.773519	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	HTTP	261	HTTP/1.1 200 OK (text/plain)
23094	173.773519	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	TCP	74	80 → 52368 [FIN, ACK] Seq=188 Ack=113 Win=64768 Len=0
23095	173.773622	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [ACK] Seq=113 Ack=189 Win=65280 Len=0
23096	173.773765	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [FIN, ACK] Seq=113 Ack=189 Win=65280 Len=0
23130	174.074054	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	74	[TCP Retransmission] 52368 → 80 [FIN, ACK] Seq=113 Ack=189 Win=652...

> Frame 23091: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface \Device\NPF_{E9:...

> Ethernet II, Src: HP_cf:c6:15 (7c:57:58:cf:c6:15), Dst: HuaweiTechno_9c:71:27 (1c:e6:39:9c:71:27)

> Internet Protocol Version 6, Src: 2001:da8:2d00:2069::4267, Dst: 2600:1417:4400:3::1731:68d6

✓ Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 112

Source Port: 52368

Destination Port: 80

[Stream index: 223]

[Stream Packet Number: 4]

> [Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 112]

Sequence Number: 1 (relative sequence number)

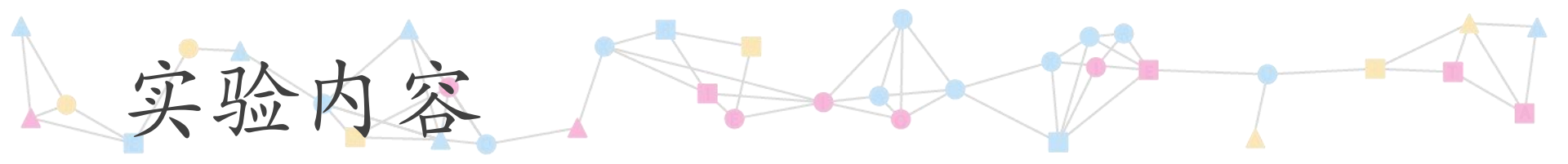
Sequence Number (raw): 3158848141

[Next Sequence Number: 113 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Destination Port (tcp.dstport), 2 byte(s)

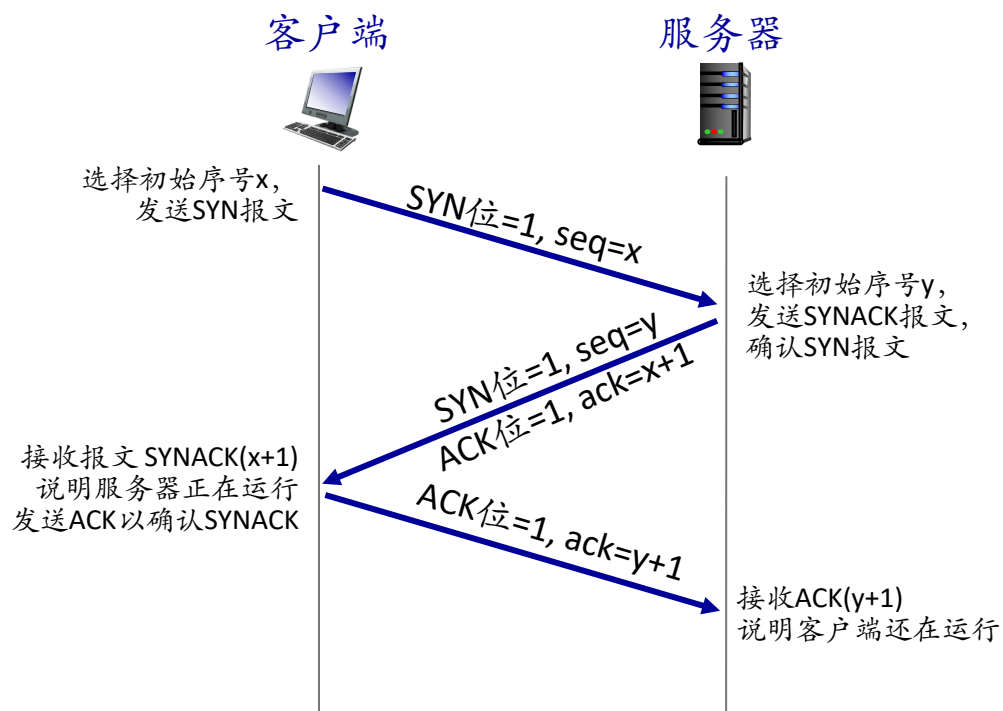
分组: 191802 · Displayed: 11 (0.0%) 配置: Default



实验内容

1. 安装学习Wireshark软件
2. 抓包与分析HTTP协议
3. 分析TCP协议
4. 分析TCP三次握手
5. 分析ICMP协议

4. TCP三次握手



4. 分析TCP三次握手

1) 分析TCP三次握手，第一次握手(SYN)。

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tcp.stream eq 223

No.	Time	Source	Destination	Protocol	Length	Info
23088	173.630693	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	86	52368 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
23089	173.703074	2600:1417:4400:3::1...	2001:da8:2d00:2069::...	TCP	86	80 → 52368 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PE...
23090	173.703152	2001:da8:2d00:2069::...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 0, Len: 0

Source Port: 52368
Destination Port: 80
[Stream index: 223]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3158848140 序号是3158848140
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
000. = Reserved: Not set
...0 = Accurate ECN: Not set
...0 = Congestion Window Reduced: Not set
...0 = ECN-Echo: Not set
...0 = Urgent: Not set
...0 = Acknowledgment: Not set
...0 = Push: Not set
...0 = Reset: Not set
...1. = Syn: Set SYN位置1
...0 = Fin: Not set

This shows the raw value of the sequence number (tcp.seq.raw) 4 byte(s)

分組: 103419 · Displayed: 11 (0.0%) 配置: Default

4. 分析TCP三次握手

2) 分析TCP三次握手，第二次握手(SYNACK)。

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tcp.stream eq 223

No.	Time	Source	Destination	Protocol	Length	Info
23088	173.630693	2001:da8:2d00:2069:...	2600:1417:4400:3::1...	TCP	86	52368 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
23089	173.703074	2600:1417:4400:3::1...	2001:da8:2d00:2069:...	TCP	86	80 → 52368 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PE...
23090	173.703152	2001:da8:2d00:2069:...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Transmission Control Protocol, Src Port: 80, Dst Port: 52368, Seq: 0, Ack: 1, Len: 0

Source Port: 80
Destination Port: 52368
[Stream index: 223]
[Stream Packet Number: 2]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2302613785
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3158848141
1000 ... = Header Length: 32 bytes (8)

Flags: 0x012 (SYN, ACK)

000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... 0... = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ..1. = Acknowledgment: Set
.... ..0... = Push: Not set
.... ..0... = Reset: Not set
>1. = Syn: Set
.... ..0 = Fin: Not set

0000 7c 57 58 cf c6 15 1c e6 39 9c 7:
0010 44 2d 00 20 06 32 26 00 14 17 4:
0020 00 00 17 31 68 d6 20 01 0d a8 2:
0030 00 00 00 00 42 67 00 50 cc 90 8:
0040 32 8d 80 12 fd 20 58 4a 00 00 0:
0050 04 02 01 03 03 07

序号是2302613785
确认号是3158848141 = SYN序号加1
ACK位置1
SYN位置1

This shows the raw value of the sequence number (tcp.seq raw), 4 byte(s) 分组: 194666 · Displayed: 11 (0.0%) 配置: Default

4. 分析TCP三次握手

3) 分析TCP三次握手，第三次握手(ACK)。

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tcp.stream eq 223

No.	Time	Source	Destination	Protocol	Length	Info
23088	173.630693	2001:da8:2d00:2069:...	2600:1417:4400:3::1...	TCP	86	52368 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
23089	173.703074	2600:1417:4400:3::1...	2001:da8:2d00:2069:...	TCP	86	80 → 52368 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PE...
23090	173.703152	2001:da8:2d00:2069:...	2600:1417:4400:3::1...	TCP	74	52368 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Transmission Control Protocol, Src Port: 52368, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

Source Port: 52368
Destination Port: 80
[Stream index: 223]
[Stream Packet Number: 3]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3158848141
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2302613786
0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK)

000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... 0... = Push: Not set
....0.. = Reset: Not set
....0. = Syn: Not set
....0 = Fin: Not set

0000 1c e6 39 9c 71 27 7c 57 58 cf c1
0010 a7 3b 00 14 06 40 20 01 0d a8 2c
0020 00 00 00 00 42 67 26 00 14 17 4d
0030 00 00 17 31 68 d6 cc 90 00 50 b1
0040 19 1a 50 10 00 ff bb b5 00 00

确认号是23026137856 = SYNACK序号加1

ACK位置1

This shows the raw value of the sequence number (tcp.seq_raw), 4 byte(s)

分组: 195577 · Displayed: 11 (0.0%) 配置: Default